

Distributed Systems and Cloud Computing

Modern cloud applications are designed using distributed systems principles. Instead of running everything on a single server, workloads are distributed across multiple services and machines.

Scalability

Scalability refers to the ability of a system to handle increasing traffic and workloads. Horizontal scaling involves adding more servers, while vertical scaling involves increasing the power of existing servers.

Load Balancing

Load balancers distribute incoming traffic across multiple backend servers. This prevents any single server from becoming overwhelmed and improves system reliability.

Microservices

Microservices architecture breaks large applications into smaller independent services. Each service is responsible for a specific business capability and can be deployed independently.

Event-Driven Architecture

In event-driven systems, services communicate asynchronously using events and queues. AWS SQS is an example of a message queue service commonly used for asynchronous workflows.

Serverless Computing

Serverless platforms such as AWS Lambda allow developers to run backend functions without managing servers. The cloud provider automatically handles scaling and infrastructure management.

Caching

Caching improves performance by storing frequently accessed data in memory. Redis is commonly used to reduce database load and improve API response times.

Database Replication

Replication creates copies of databases across multiple servers to improve reliability and availability. If one database instance fails, another replica can continue serving requests.

Monitoring and Observability

Production systems require monitoring tools to track logs, metrics, and failures. CloudWatch and Grafana are common tools used to monitor cloud infrastructure.

Conclusion

Distributed systems and cloud technologies enable scalable, fault-tolerant, and highly available applications capable of serving millions of users.